

Muhammad Abdul Rehman

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in Data Science, with hands-on experience building AI-powered applications using Python, FastAPI, LangChain, TensorFlow, React, and modern LLM frameworks. Experienced in developing production-oriented RAG systems, local LLM deployments, multimodal AI solutions, REST APIs, and intelligent automation platforms. Passionate about AI Engineering, Machine Learning, NLP, Computer Vision, and Generative AI.

TECHNICAL SKILLS

- **AI / ML:** Generative AI, RAG, LLMs, NLP, Computer Vision, Machine Learning, Prompt Engineering, LLM Evaluation
- **Frameworks:** LangChain, Hugging Face, Groq, TensorFlow, Scikit-learn, OpenCV, LoRA, QLoRA
- **Programming & Web:** Python, JavaScript, FastAPI, Flask, REST APIs, React.js, Streamlit
- **Data / Databases:** Pandas, NumPy, BeautifulSoup, Scrapy, ETL Pipelines, Power BI, PostgreSQL, MongoDB, Pinecone, Firebase
- **Cloud / DevOps:** AWS (EC2), Azure AI Services, Docker, GitHub, Swagger, ReDoc

PROFESSIONAL EXPERIENCE

Data Scraping Intern | Programmers Force | Remote

July 2025 – Feb 2026

- Scraped and processed data from **2,000+ websites across 30+ countries** using a custom Chrome extension, configuring CSS/HTML tag selectors via browser DevTools (inspect mode) to extract structured data fields.
- Built entity-filtering pipelines to retain only records with verified person names; performed QA and data preprocessing, identifying malformed entries and escalating scraping failures with detailed error reports.

Full Stack Developer | Freelance | Remote

Oct 2025 – Present

- **PTC Eightfold Autofill Chrome Extension:** Built a Chrome extension (Manifest V3, chrome.storage.local, content/background scripts) that auto-fills job application form text fields, dropdowns, checkboxes, dates, and resume/cover-letter file uploads from saved or JSON-imported candidate data, eliminating repetitive manual entry.
- **Event Management System:** Built a REST API and CLI (Python, FastAPI, SQLAlchemy, SQLite) for a client to manage events and registrations, with full CRUD operations, concurrency-safe booking logic using SQLite WAL mode and locking to prevent overbooking, and a Rich-based interactive CLI as an alternative client to the API.
- **Data Automation & Analytics:** Designed advanced financial and expense tracking sheets for corporate clients utilizing complex Excel formulas, lookup functions, pivot tables, and data validation to automate manual budgeting workflows.

PROJECTS

• **IntraView AI (AI-Based Interview System)** | Final Year Project

Sept 2025 – May 2026

Python, React, TypeScript, FastAPI, REST APIs, MongoDB, TensorFlow, Hugging Face, OpenCV, Docker

- Co-developed a **dual-role full-stack platform** (candidate + admin) utilizing FastAPI, React, TypeScript, and MongoDB with JWT authentication, role-based access control, and SMTP-based OTP password reset.
- Built a **2.5M-question interview bank** (coding, behavioural, system design) with difficulty controls and a **real-time Gemini API fallback** that generates questions on demand when no DB match is found.
- Built OpenCV + MediaPipe pipelines for real-time facial expression analysis; browser-based **Docker sandbox** for automated test-case grading; admin dashboard with live session monitoring and Swagger/ReDoc API docs.

• **Privacy Medical Chatbot**

Feb 2026 – Apr 2026

Python, Meta Llama 3.1 8B, LangChain, Hugging Face, Pinecone, RAG, NLP, Flask, AWS EC2

- Deployed Meta Llama 3.1 8B locally via **Q4_K_M 4-bit quantization** (~4.58 GB), achieving **zero third-party API dependency** and full medical data privacy with a 2,048-token context window.
- Built RAG pipeline using LangChain, MiniLM-L6-v2 (384-dim embeddings), and Pinecone serverless search with 500-char PDF chunks; **top-2 retrieval in under 1 second** and custom hallucination reduction prompt template that strictly grounds LLM responses in retrieved context with source attribution, with 2 to 5 second end-to-end response times and supporting concurrent multi-user access.

• **AI Face Shape Detector & Stylist**

Sep 2025 – Dec 2025

Python, TensorFlow, EfficientNetV2, CNN, OpenCV, MediaPipe, DeepFace, FastAPI

- Trained EfficientNetV2 via 2-stage transfer learning (frozen backbone then top-30% fine-tune with early stopping), improving validation accuracy from **60% to 83.4%**; hybrid classifier combining MediaPipe's **478-point landmarks** with confidence scores.
- Integrated DeepFace for **gender detection**, serving personalized recommendations (beard/hairstyle/glasses for males; makeup/hairstyle/glasses for females) via FastAPI, supporting live webcam and file upload.

EDUCATION

COMSATS University Islamabad | Pakistan

Sep 2022 – Jun 2026

BS Computer Science – Specialization: Data Science

CERTIFICATIONS

Generative AI with Large Language Models | DeepLearning.AI & AWS | Coursera

2025

ACTIVITIES

- Served as **Registration Club Co-Lead for ACM COMSATS**, coordinating student registrations for academic and tech events.
- Organized and managed events, including **VisioSpark Episode 20** and **Student Week**.